

Finance Operations Automation Project

Reconciliation · Accounts Payable · Accounts Receivable · Compliance · Risk · Procurement · Forecasting · Reporting Automation

Project Documentation

1. Business Problem

A mid-size company's finance operations team currently relies on manual, disconnected processes to reconcile transactions, track payables/receivables, monitor compliance, and assess vendor/transaction risk. This leads to delayed exception detection, inconsistent reporting, and no standardized way to catch fraud-adjacent patterns — duplicate payments, segregation-of-duty violations, unmatched receipts — before they compound.

This project answers: “How can a finance operations team use Excel to detect discrepancies, monitor payables and receivables, assess compliance and risk, and generate accurate recurring reports — with minimal manual effort?”

2. Dataset

Source file: Finance_Audit_Raw_Data_1.xlsx, covering January 2025 through mid-2026 across nine linked tables.

Table	Rows	Purpose	Key
Vendor_Master	40	Vendor details, onboarding dates, payment terms	Vendor_ID
Customer_Master	30	Customer details (AR side)	Customer_ID
Purchase_Orders	850	PO amounts, requester/approver	PO_ID
Goods_Receipt_Notes	790	Confirms goods actually received	PO_ID
Vendor_Invoices_AP	881	Invoice amounts vs PO	Invoice_ID / PO_ID
Payment_Log_AP	752	Actual AP payments made	Payment_ID / Invoice_ID
Bank_Statement	782	Bank-side transaction record	Bank_Txn_ID
Customer_Invoices_AR	600	Billing data	AR_Invoice_ID
Cash_Receipts_AR	502	Customer payments received	Receipt_ID / AR_Invoice_ID

3. Workbook Architecture

Everything lives in one workbook, Finance_Automation_Workbook.xlsx, so every module recalculates together with no external file links (a deliberate choice for the older, non-365 Excel build this was designed for — no Power Query, no XLOOKUP/dynamic arrays). Tabs are color-coded by function and every calculation is a live formula, not a pasted value, so the whole thing re-derives itself the moment the raw data or the Assumptions tab changes.

Tab group	Tabs	Color
Navigation	Cover, Assumptions	Navy
Raw Data	9 source tables (untouched)	Grey
Reconciliation	3Way_Match, Bank_Reconciliation	Blue
Accounts Payable	AP_Aging, Duplicate_Payments, AP_Payment_Detail, DPO_Summary	Orange
Accounts Receivable	AR_Aging, AR_Partial_Unpaid, AR_Payment_Detail, DSO_Summary	Green
Compliance	SoD_Check, RCM	Purple
Risk	Risk_Scoring	Red
Procurement	Vendor_Spend, New_Vendor_Risk	Gold

Forecasting	Cash_Outflow_Forecast	Light blue
Reporting	Report_Weekly/Monthly/Quarterly/Yearly, KPI_Dashboard	Navy

4. Methodology by Module

4.1 Reconciliation

- 3-Way Match (tab: 3Way_Match) — for every PO, sums the matching Goods_Receipt_Notes and Vendor_Invoices_AP amounts (SUMIFS on PO_ID), then flags an EXCEPTION when goods were invoiced without a receipt, or when the PO-vs-GRN or PO-vs-Invoice variance exceeds the tolerance set in Assumptions (default \$1).
- Bank Reconciliation (tab: Bank_Reconciliation) — Part A matches every AP payment to the bank statement by Payment_ID (INDEX/MATCH) and flags Unmatched or amount Mismatch. Part B works the other direction, flagging bank lines with no payment reference as “Unexplained” (fees, timing, or other bank-only activity).

4.2 Accounts Payable

- AP Aging (tab: AP_Aging) — Outstanding = Invoice Amount minus payments received-to-date (SUMIFS from Payment_Log_AP), bucketed 0-30 / 31-60 / 61-90 / 90+ days as of the Assumptions As-Of date.
- Duplicate Payment Detection (tab: Duplicate_Payments) — COUNTIF of Payment_Log_AP by Invoice_ID; any invoice with more than one payment is flagged for review, along with any invoice paid more than its billed amount.
- DPO (tab: DPO_Summary) — headline $DPO = \text{Ending AP} \div \text{Total AP Invoiced} \times \text{days in the data period}$ (the standard formula), plus an actual “average days-to-pay” trend by month, computed from a payment-to-invoice-date match in the AP_Payment_Detail helper tab.

4.3 Accounts Receivable

- AR Aging (tab: AR_Aging) mirrors AP Aging, using Customer_Invoices_AR and Cash_Receipts_AR.
- Partial/Unpaid Tracking (tab: AR_Partial_Unpaid) flags short-paid invoices as Medium risk and invoices unpaid past 90 days as High risk.
- DSO (tab: DSO_Summary) mirrors the DPO methodology on the AR side, using the AR_Payment_Detail helper tab.

4.4 Compliance

- Segregation of Duties (tab: SoD_Check) — flags every PO where Requested_By equals Approved_By, a classic fraud-control break.
- Risk & Control Matrix (tab: RCM) — a real internal-audit artifact: for each key process it states the risk, the control, the test performed, and pulls a live Result and Pass/Fail straight from the modules above (e.g. Bank Reconciliation control fails automatically if any unmatched items exist).

4.5 Risk Assessment

- Weighted Risk Scoring (tab: Risk_Scoring), an AML-style score applied to every AP payment:

Factor	Points	Trigger
Large amount	40	$\geq$ Assumptions large-transaction threshold (\$10,000 default)
Elevated amount	20	$\geq$ Assumptions elevated threshold (\$5,000 default), below Large
Round-dollar amount	10–20	Payment amount divisible by 100 or 500
New vendor	25	Vendor onboarded within the Assumptions new-vendor window (180 days default) of the payment date
Higher-risk geography	15	Vendor country is on the illustrative watch-list in Assumptions

Scores of 60+ are Risk_Level “High”, 30–59 “Medium”, below 30 “Low”. The geography list and thresholds are placeholders — replace them with your actual AML policy in the Assumptions tab; the note there flags this explicitly.

4.6 Procurement

- Vendor Spend Analysis (tab: Vendor_Spend) — total PO spend per vendor, ranked and accumulated (Pareto) to show which vendors drive 80% of spend.
- New Vendor Risk Flagging (tab: New_Vendor_Risk) — flags vendors onboarded within the Assumptions window whose total spend is more than 2x the average vendor's spend.

4.7 Forecasting

- Cash Outflow / AP Forecasting (tab: Cash_Outflow_Forecast) — 19 months of actual AP payment totals, projected 3 months forward with the classic FORECAST() linear-regression function (chosen instead of FORECAST.LINEAR/FORECAST.ETS for compatibility with older, non-365 Excel).

5. Reporting Automation Layer

Report_Weekly, Report_Monthly, Report_Quarterly and Report_Yearly are built as formula-driven summary tables grouped by period (the pivot-style rollup the brief calls for), each showing: AP payments, AR receipts, 3-way match exceptions, duplicate-payment flags, SoD violations, high-risk transactions and new POs raised in that period. KPI_Dashboard is the one-page snapshot — headline KPIs plus AP/AR aging-mix pie charts, DPO/DSO trend, top vendors and a live Risk & Control Matrix status panel — that the VBA macro exports to PDF.

Note on PivotTables: this workbook's summary tables are formula-based (SUMIFS/COUNTIFS) so they work identically in the older Excel build this project targets and recalc automatically with everything else. If you prefer native Excel PivotTables on top of the module tabs (e.g. 3Way_Match, Risk_Scoring), the data is already structured as clean tables — select any module tab and use Insert > PivotTable; right-click > Refresh (or the macro's RefreshAllReports) will keep those in sync too.

6. Automation — VBA Macro

Finance_Automation_Macros.bas contains three macros (see the separate How-To-Run Guide for installation steps). The automation scope is deliberately reports-only: it refreshes every module and produces a dated PDF snapshot saved to a folder next to the workbook. There is no automated email step — an earlier version explored an Outlook/Gmail-based auto-send, but this added an external dependency (mail client or SMTP access) without being essential to the core brief, so it was removed in favor of a simpler, fully self-contained macro that always works regardless of what mail setup is available on the machine.

Macro	What it does
RefreshAllReports	Forces a full recalculation of every formula and refreshes any PivotTables added on top of the data.
ExportReportsToPDF	Exports Report_Weekly, Report_Monthly, Report_Quarterly, Report_Yearly and KPI_Dashboard together as one dated PDF, saved to a Reports_Exported folder created next to the workbook.
RunAllReports	Runs both macros above in sequence — the one-click button macro — then shows a confirmation with the exact file path of the PDF it just created.

The exported PDF is a normal file on disk, so it can still be attached to an email, uploaded, or shared manually with no extra steps.

Scheduling: VBA cannot run while Excel is closed. For a genuine “every Monday at 9am” automated report generation, pair this workbook with Windows Task Scheduler, which opens the file on schedule and triggers the macro (an Auto_Open template is included in the module, disabled by default). This is the standard corporate pattern for unattended Excel automation and is documented step-by-step in the How-To-Run Guide.

7. Key Assumptions

All editable on the Assumptions tab (yellow cells):

- Reporting As-Of Date: 2026-07-31 (source data runs through mid-2026).
- Match tolerance for 3-way match / bank reconciliation: \$1.00.
- Large / Elevated transaction thresholds for risk scoring: \$10,000 / \$5,000 (illustrative — align to your AML policy).
- New-vendor window: 180 days.
- Higher-risk country list: India, UAE, Singapore (illustrative placeholder, not a real AML geography list).
- Aging buckets: 0-30 / 31-60 / 61-90 / 90+ days, used consistently for AP and AR.

8. Known Limitations

- Risk-scoring weights and the geography list are illustrative starting points, not a validated AML model — they should be reviewed and calibrated against your company's actual policy before any operational use.
- DPO/DSO trend uses the payment date matched to the first invoice reference found; an invoice paid across multiple partial payments is captured once per payment, not netted, which is disclosed on the DPO/DSO tabs.
- The reporting tabs are formula-built summaries rather than native Excel PivotTable objects, by design, for compatibility with the older Excel build this project targets (see Section 5).
- This is built on a static, one-time raw-data extract. In production the raw-data tabs would be replaced by a live data connection (ODBC/database import) refreshed on the same schedule as the macro.